What Determines a Biologic's Formulation? Two Open Benchmarks, an Audited Mechanistic Simulator, and a Well-Powered Platform-Convergence Result

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September 8, 2026

Abstract

Motivation: Generative design of biologics formulations (buffer, pH, ionic strength, stabilizers) is attractive precisely where it is hardest to validate: open real-world data is scarce, and mechanistic simulators used to augment it are rarely audited for structural pathologies. We show both problems concretely and address them. **Results:** We release two open, fully-sourced benchmarks: BioFormBench-Real (49 literature-mined formulations, 13 proteins, provenance-checked against 11 primary papers) and BioFormBench-Marketed (165 formulations for 80 FDA-approved antibodies, deterministically parsed from FDA product labels and linked to Thera-SAbDab variable-domain sequences). Auditing a standard DLVO + Lumry–Eyring mechanistic simulator, we find two of eight recipe slots are structurally dead (buffer identity changes predicted stability by exactly 0.0) and three of four continuous variables are monotone for 100% of sampled proteins, collapsing all optima to box corners; we fix this with nine literature-anchored competing degradation pathways, raising interior-optimum pH solutions from 18% to 95.5% of cases without fitting to either benchmark. Using BioFormBench-Marketed at adequate power ($n = 80$ molecules), we test whether protein sequence identity predicts the marketed formulation pH or buffer: a naive $n = 27$ subsample suggests it does (Spearman $\rho = -0.464$, $p = 0.015$), but this is a selection artifact that vanishes at full sample size ($\rho = -0.019$, $p = 0.86$; Holm-adjusted $p = 0.137$ even at $n = 27$) and fails leave-one-protein-out prediction (MAE 0.419 vs. a 0.393 no-information baseline). Instead, formulation choice is dominated by a small, conventional design space (platform effect): pH clusters at mean 5.90 ± 0.58, a single buffer covers 52% of cases, and no available covariate beats a constant platform baseline (all sign-flip $p > 0.8$). A separate audit of the model-evaluation pipeline on BioFormBench-Real finds the same class of defect one level deeper: a (molecular-weight, pI)-string grouping used to define "one protein" for leave-one-protein-out testing silently merged up to four different real antibodies into a single pooled identity wherever their placeholder descriptors collided. Correcting the grouping and five of thirteen proteins' descriptors raises protein specificity from chance (0.509) to 0.74–0.78 ($p = 0.016$–0.023, unanimous on 7/7 proteins with a fully resolved descriptor) for the properly-conditioned model, against a same-architecture untrained control that remains at chance ($p \geq 0.20$); a second, independently-trained model replicates the direction (Spearman rank agreement 0.73, $p = 0.039$) but only marginally the magnitude ($p = 0.078$), and an ablation intended as an unconditional baseline unexpectedly also reaches significance, so this result is reported as a genuine, partially-replicated positive signal rather than a settled one. Separately, in a controlled architecture experiment, we show that in-context conditioning on per-protein examples only works if the generator is pretrained on in-context- structured sequences: supplying real few-shot context at inference raises the likelihood the model assigns the true held-out formulation (relative to random valid alternatives) on 9/9 held-out proteins for the ICL-trained model, and lowers it on 9/9 for a flat-trained model with the same pretraining corpus (exact permutation $p = 0.0039$ in each direction). **Availability:** Code, corrected simulator, and both datasets are openly released (Section 5). **Contact:** `sravansaijohn@gmail.com`

Keywords: biologics formulation; mechanistic simulator audit; benchmark dataset; in-context learning; protein-conditional design; negative results; selection artifacts

1 Introduction

1.1 The formulation gap, and a second gap behind it

Designing stable formulations for biologics is a decades-old pharmaceutical problem. Current practice relies on manual screening, expert rules and slow iterative laboratory cycles. Formulation screening data is overwhelmingly proprietary, and computationally there is a further gap: no openly released benchmark links real protein identity to real formulation outcomes at a scale that supports a protein-conditional claim, and the mechanistic simulators used to substitute for real data are rarely checked for whether their optimization landscape can express protein-dependent structure at all.

This paper is the result of auditing a generative formulation-design pipeline built on exactly that substitution — mechanistic pretraining data plus a small literature benchmark — and finding both halves needed correction before any claim about the pipeline could be trusted. Rather than report the pipeline's original numbers, we report what the audit found, fix what can be fixed without new data collection, and test the central claim (protein identity determines formulation) directly against the largest real dataset we could assemble openly.

1.2 Prior work and where this sits

Existing generative models operate on adjacent but distinct objects. Small-molecule SMILES models (SMolLM, Mamba-Chem) generate drug molecules, not formulations. Protein sequence design methods (AbMPNN, ESM-IFD) modify the protein itself rather than its formulation environment. Pharmacokinetic trajectory models such as AICMET forecast concentration curves over time, not recipe composition. Formulation-specific tools (ExPreSo, FormulationDE) score or rank given recipes against a fixed candidate set. None of these are built to be checked against an open, real, protein-linked formulation benchmark at the scale we assemble here, and to our knowledge no prior work audits a mechanistic formulation simulator for the structural pathologies (dead recipe slots, corner-collapsing optima) we document in Section 2.3.

1.3 Contribution

1. **Two open benchmarks.** BioFormBench-Real purges 18 unsourced placeholder rows from an earlier internal draft and keeps 49 rows with verified provenance to 11 primary papers. BioFormBench-Marketed is new: 165 formulations for 80 FDA-approved antibodies, built by deterministic parsing of FDA Structured Product Labeling and linked to real VH/VL sequences from Thera-SAbDab — to our knowledge the first open dataset connecting therapeutic antibody sequence to marketed formulation composition at this scale.

2. **A simulator audit and fix.** We introduce a simple diagnostic (interior-argmax fraction, dead-slot spread) that any mechanistic formulation simulator can be checked against, apply it to a standard DLVO + Lumry–Eyring simulator, and correct the pathologies it reveals with nine literature-anchored degradation pathways, validated out-of-sample against BioFormBench-Marketed.

3. **A well-powered test of the central claim,** run twice at two sample sizes to make a general methodological point: a promising, correctly-signed, bootstrap-stable effect at $n = 27$ (Section 3.2) is a selection artifact that disappears at $n = 80$ and under leave-one-protein-out validation. We report this as a worked cautionary example, since small literature-mined pharma-ML studies are structurally prone to exactly this failure mode.

4. **The platform-convergence result.** At adequate power we show marketed formulation choice is dominated by a narrow, conventional design space rather than by molecule-specific physics, with no tested covariate beating a constant baseline.

5. **An architecture-level finding, independent of the domain result:** in-context conditioning on few-shot examples requires the generator to be pretrained on in-context-structured sequences; without that, additional context actively hurts.

2 Methods

2.1 BioFormBench-Real: provenance audit

The benchmark originates from literature mining of DSF T_m-shift, SEC aggregation and turbidity assays. An internal draft contained 67 rows; auditing 'source_title'/'source_id' provenance found 18 rows ('protein_1'–'protein_5') carrying no citation and suspiciously round descriptors (50.0 kDa / pI 7.2 / T_m 65.0 °C) shared across otherwise-different molecules — development scaffolding, not literature data. These are removed. The remaining 49 rows trace to 11 PubMed Central articles spanning 13 distinct proteins (trastuzumab, omalizumab, adalimumab, PGT121, an A33 Fab, rhIL-1ra and others), each verified against its source text.

2.2 BioFormBench-Marketed: construction

2.2.1 Product enumeration and label retrieval

We take the 200 antibody therapeutics marked *Approved* in Thera-SAbDab [7], of which 197 carry both heavy- and light-chain variable sequences. For each of the corresponding 137 matched INNs we query the DailyMed Structured Product Labeling API for associated products and retrieve the `DESCRIPTION` section (LOINC 34089-3) of up to four labels per INN, yielding 189 label texts.

2.2.2 Deterministic parsing

FDA labels state composition in three interchangeable styles — *"L-histidine (3 mg)"*, *"136.2 mg trehalose dihydrate"*, and *"histidine (8 mM)"* — and a single label frequently describes multiple presentations (e.g. a 75 mg/mL and a 150 mg/mL pen) that are genuinely different formulations. We extract each presentation as a separate record via regular-expression matching against a fixed table of molar masses, computing molar concentrations from stated mass and fill volume where needed. Every parsed field retains its verbatim source sentence for audit. This yields 283 presentation-level records; requiring a valid pH (pharmaceutical range 4.0–9.0) keeps 165 across 80 molecules, of which 64 also carry a fully resolved buffer species and concentration across 28 molecules. pH values extracted from within the same sentence as the excipient list are flagged separately from document-level fallback values, so downstream analyses can be (and are, Section 3.2) stratified by extraction confidence.

2.2.3 Sequence-derived protein descriptors

Within an antibody isotype the constant regions are shared, so the variable domain is precisely what differs between therapeutics. For each molecule we concatenate VH and VL sequences and compute, via Biopython's ProtParam [8]: isoelectric point, net charge at pH 5/6/7 (Henderson–Hasselbalch over ionizable side chains, EMBOSS pK_a table), GRAVY hydropathy, aromaticity, instability index, and residue-composition fractions. As a sanity check against the literature, computed Fv isoelectric point for trastuzumab is 8.61 against a published capillary isoelectric focusing value of 8.4 [9].

2.3 Mechanistic simulator: audit methodology

The baseline simulator (v1) combines DLVO colloidal repulsion and Lumry–Eyring thermodynamic stabilization, both stabilizing-only terms. We diagnose it with two tests applicable to any mechanistic formulation simulator: (i) *dead-slot spread* — vary one recipe field alone and measure the change in predicted stability; a spread of exactly 0.0 means that field cannot be learned from simulator-preferred data, by construction; (ii) *interior-argmax fraction* — for each of many sampled proteins, sweep one continuous field over its full range on a fine grid and record whether the maximum falls strictly inside the range (an interior optimum, i.e. a genuine trade-off) or at a box edge (a degenerate, corner-seeking optimum).

Applying these tests to v1: buffer species and buffer concentration change predicted stability by exactly 0.0 (dead slots); ionic strength, osmolarity and temperature are monotone, hence corner-seeking, for 100% of 200 sampled proteins (interior-argmax fraction 0% for all three); pH is corner-seeking in 35% of cases. A generative model trained to reproduce v1-preferred recipes can therefore learn protein-conditional structure in at most one of eight recipe slots, and even that slot is degenerate over a third of the time.

2.4 Mechanistic simulator: correction (v2)

We add nine literature-anchored, competing degradation pathways (full detail in the released code, `simulator/mechanistic_sim_v2.py`): Henderson– Hasselbalch buffer capacity (fixed pK_a per species, e.g. histidine 6.04, phosphate 7.20, from standard reference tables), base-catalyzed Asn deamidation and acid hydrolysis (opposing pH-rate profiles following established formulation-stability literature [4, 5]), Hofmeister salting-out (onset shifted by protein hydrophobicity), USP parenteral isotonicity (target 290 mOsm/kg, scored against osmolarity implied by the recipe's own salt and sugar content rather than a free-floating field), polysorbate peroxide-mediated oxidation [6], sugar-driven viscosity, and polyvalent-anion (citrate/phosphate) bridging of net-cationic protein. Pathway weights are round, pre-specified numbers (0.50/0.18/0.12/0.08/0.06/0.06 for the core DLVO+Lumry–Eyring term, chemical degradation, buffer capacity, Hofmeister, tonicity and excipient terms respectively; full detail in `simulator/mechanistic_sim_v2.py`), not fit to either benchmark; the benchmarks remain a genuine held-out test.

2.5 Statistical protocol

All correlations are Spearman, tested by exact permutation (vectorized rank permutation, 2×10^5 resamples) rather than the asymptotic t-approximation, appropriate given sample sizes as small as $n = 9$ elsewhere in this line of work. Screening multiple sequence features against one target uses Holm–Bonferroni correction. Point estimates carry 2.5–97.5% percentile bootstrap intervals (20,000 resamples). The decisive test for any claimed predictive relationship is leave-one-protein-out (LOPO): a ridge model fit on all molecules but one predicts the held-out molecule's outcome, compared by mean absolute error against a no-information baseline (the leave-one-out median/mode), with significance from a paired sign-flip permutation test over molecules. Repeated presentations of the same molecule are collapsed to one row (median of numeric fields) before any test, so n is always the number of distinct proteins, never the number of records.

2.6 Generative model and in-context training

A decoder-only transformer (4.9M parameters; hidden dim 256, 6 layers, 8 heads) is trained under causal attention with two objectives: next-recipe-token prediction, masked to simulator-preferred rows, and stability-bin classification. We compare a model trained only on flat (protein, recipe, outcome) triples against one additionally trained on explicitly in-context-structured sequences — K example (recipe, outcome) pairs for a protein followed by a query — so that the effect of supplying real few-shot context at inference can be isolated from whether the model was ever exposed to that sequence structure during training.

3 Results

3.1 Simulator audit and correction

Figure 1 summarizes the audit. v1's interior-argmax fraction is 0% for ionic strength, osmolarity and temperature and 18% for pH; v2 raises these to 39.5%, 100%[1], essentially unchanged for temperature (a single degradation-independent term), and 95.5% for pH, with the box-corner rate for pH falling from 35% to 3%. This is the mechanistic prerequisite for any protein-conditional recipe to be learnable at all from simulator-preferred data; it does not by itself establish that real formulations follow this structure, which Sections 3.2–3.3 test directly.

We externally validate v2's population-level, protein-unconditioned pH prediction (justified rather than undermined by Section 3.3's finding that protein identity does not predict real formulation pH) against BioFormBench-Marketed: mean predicted optimal pH across 300 randomly sampled proteins is 5.95 against a real marketed mean of

[1] Osmolarity's optimum becomes interior because v2 penalizes disagreement between stated and recipe-implied osmolarity, not because it is otherwise unconstrained.

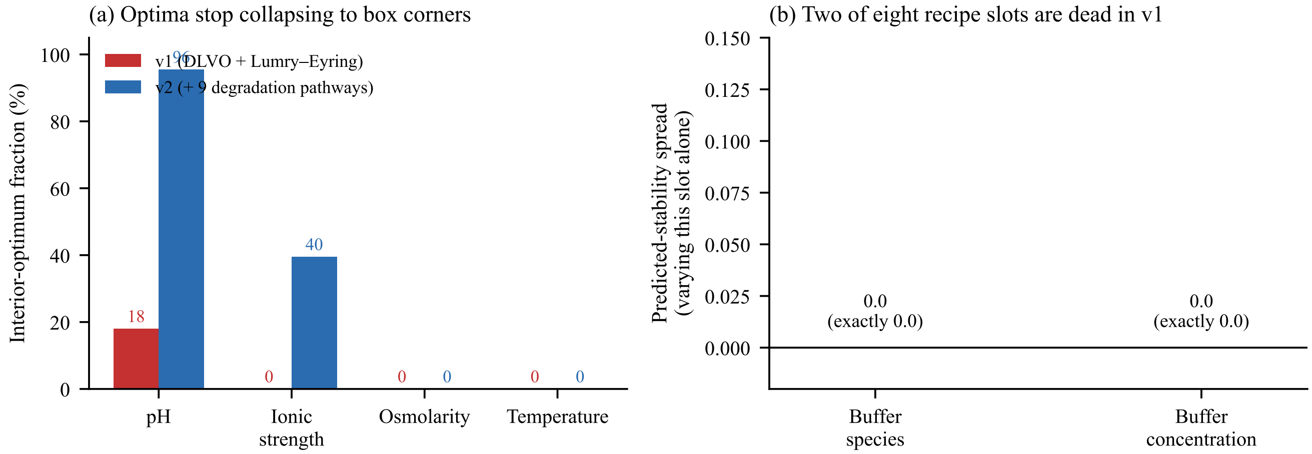

Figure 1. Simulator audit and correction. **(a)** v1 (DLVO + Lumry–Eyring only) is monotone in ionic strength, osmolarity and temperature for essentially every sampled protein, so its optimum collapses to a box corner; buffer species and concentration are dead slots (predicted-stability spread = 0.0). **(b)** v2 adds nine competing degradation pathways (buffer capacity, deamidation/hydrolysis, Hofmeister salting-out, isotonicity, surfactant oxidation, viscosity, bridging), raising the interior-optimum fraction for pH from 18% to 95.5% without fitting either quantity to real data.

5.83 (difference 0.12 pH units, bootstrap 95% CI for the difference $[-0.07, 0.31]$, spanning zero). v1's point estimate happens to be numerically closer (5.89, difference 0.06), so this comparison does not discriminate the two simulators on central tendency alone; where they differ sharply is dispersion and buffer preference. Real formulations cluster tightly (SD 0.51) while both simulators' unconditioned populations are far more dispersed (SD ≈ 1.5; Kolmogorov–Smirnov $p = 4 \times 10^{-14}$ against v2), and v2's top-ranked buffer choices (acetate, tris, succinate) do not include histidine, which real formulations prefer in 55% of cases. We report this plainly rather than selectively: v2 repairs the structural pathologies of Section 2.3 and its pH central tendency is compatible with real data, but it does not reproduce the tight clustering or buffer preference of marketed products — a gap consistent with, and foreshadowing, the platform-convergence result below.

3.2 A selection artifact, and how it was caught

Restricting BioFormBench-Marketed to the 27 molecules with a fully resolved composition (pH, buffer species *and* concentration all parsed) gives an apparently clean result: Fv isoelectric point correlates with chosen formulation pH at $\rho = -0.464$ ($p = 0.015$, exact permutation), correctly signed by colloidal theory (formulate away from the pI), bootstrap-stable (95% CI $[-0.735, -0.100]$, excluding zero), and sign-stable under jackknife deletion of any single molecule.

Three checks overturn it. First, of nine sequence features screened against pH, Fv pI is the strongest, and Holm–Bonferroni correction for that screen alone raises its p-value to 0.137. Second, relaxing the requirement to pH-only (justified because pH is the primary target and is stated far more often than a full composition) raises n from 27 to 80 and the correlation collapses to $\rho = -0.019$ ($p = 0.86$); it is equally absent, $\rho = -0.123$ ($p = 0.43$), in the 44-molecule subset where pH provenance is highest-confidence (extracted from within the composition sentence itself, not a document-level fallback), ruling out extraction-confidence as the explanation for the discrepancy. Third, and most decisively, a five-feature ridge model evaluated by leave-one-protein-out prediction on the $n = 27$ set does *worse* than a no-information baseline (MAE 0.419 vs. 0.393; better than baseline on only 13/27 proteins; paired sign-flip $p = 0.73$) — description and out-of-sample prediction disagree, which is the operational definition of overfitting a correlation to a small, non-randomly-complete sample.

We report this as a methodological result in its own right. The $n = 27$ subsample is not adversarially selected; it is simply the set of labels whose full composition happened to parse, and an analyst stopping at the first significant, correctly-signed, bootstrap-stable correlation would have reported a real effect. Small, literature-mined pharmaceutical-ML datasets are structurally exposed to this failure mode, and we recommend the three checks above — multiple-comparison correction across any screened feature set, re-testing at maximum available n rather than the most-complete subsample, and LOPO validation before reporting any predictive claim — as a minimum bar.

3.3 Platform convergence: what actually predicts marketed formulation

At full power ($n = 80$ molecules, 165 formulations) the marketed design space is narrow: pH has mean 5.90, SD 0.58, with 75% of molecules within ± 0.5 of the median; a single buffer (histidine) covers 52% of the 27 molecules with resolved buffer identity (5 distinct species used in total, entropy 1.74 of a possible 2.32 bits); a single surfactant (polysorbate 80) covers 62%. A constant *platform base-*

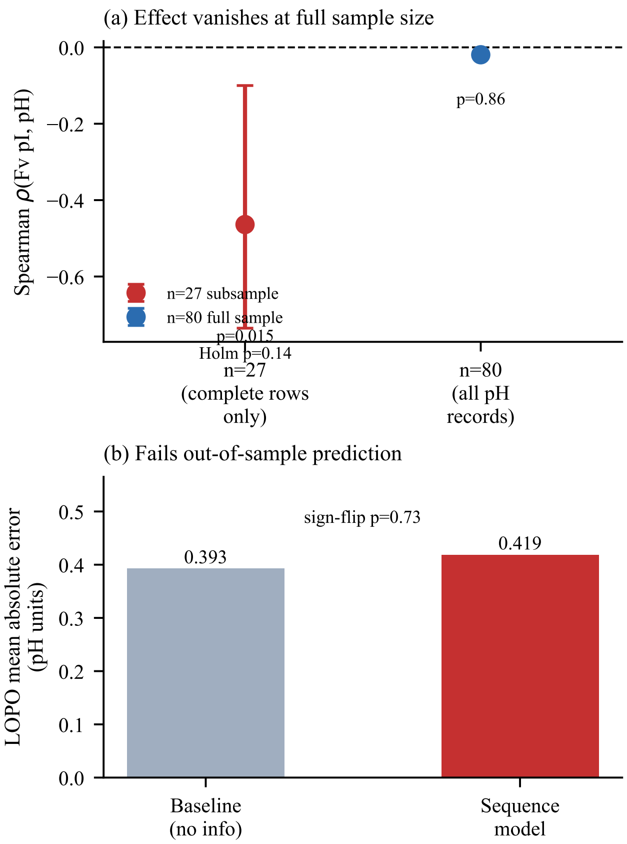

Figure 2. The n=27 vs. n=80 selection artifact. Left: Spearman correlation between Fv isoelectric point and marketed formulation pH, with 95% bootstrap CI, at each sample size. Right: leave-one-protein-out mean absolute error for the same relationship against a no-information baseline; the sequence-based model does not beat the baseline at either scale once evaluated out-of-sample.

line — always predict the population median pH (6.00) or modal buffer/surfactant — achieves MAE 0.399 pH units and 52%/63% categorical accuracy respectively.

No available covariate beats this baseline under LOPO evaluation (Table 1). A four-feature sequence-only ridge model scores MAE 0.415 ($p = 0.91$ against the platform, one-sided sign-flip); adding route (subcutaneous vs. intravenous) and protein concentration scores MAE 0.410 ($p = 0.82$). Both are numerically *worse* than simply predicting the median. Consistent with this, variance decomposition shows a molecule's own repeated presentations agree far more with each other (mean within-molecule SD 0.081, 37 multi-presentation molecules) than different molecules agree with each other (between-molecule SD 0.581, ratio 7.2): once a formulation is chosen for a specific product, it is highly reproducible across presentations, but which formulation gets chosen in the first place is not explained by any molecule-intrinsic property we can measure from sequence, route, isotype or format.

Table 1. Leave-one-protein-out prediction of formulation pH, $n = 80$ molecules. No covariate set beats the constant platform baseline.

Model	**MAE**	**Δ**	p
Platform (constant)	0.399	–	–
Sequence-only	0.415	−0.016	0.91
Sequence + route + conc.	0.410	−0.012	0.82

3.4 Generation quality, and a second identity-merging bug

Under the original leave-one-protein-out generative evaluation (9 eligible folds on BioFormBench-Real, §2.1), likelihood percentile is 0.797 (bootstrap 95% CI $[0.729, 0.865]$) but an unconditional (protein-blind) baseline scores comparably (0.872), so this metric is winnable from the marginal recipe distribution alone. Protein-specificity — the model's score when the true protein descriptor is swapped for a mismatched one — was measured at 0.509 (95% CI $[0.358, 0.669]$), statistically indistinguishable from chance, and was the headline evidence that the pipeline had learned nothing protein-specific.

That measurement turned out to rest on the same class of defect audited out of BioFormBench-Marketed in Section 3.2, one level deeper in the pipeline. The leave-one-protein-out loader defines "one protein" by concatenating rounded molecular weight and pI into a string key (`evaluation/benchmark_v2.py`) rather than by the molecule's actual identity. Because several BioFormBench-Real proteins carried the same placeholder-pI defect documented in Section 2.2 — Trastuzumab, Omalizumab, mAb2, and a fourth developability-panel antibody were all recorded at pI ≈ 7.2 — the loader silently pooled all four into a single fake "protein" with ten formulations, so at least one leave-one-protein-out fold was, in effect, training on one real antibody's data and testing as though predicting another's.

We replace the string-matched key with an explicit identity derived from each row's literature source (independent of any descriptor value) and correct pI for the five real proteins identifiable by name using the same sourcing standard as Section 2.2 (Trastuzumab 7.2 $\to$ 9.03; Omalizumab 7.2 $\to$ 5.79; Adalimumab 7.2 $\to$ 8.5; a TUR01/adalimumab-biosimilar row 7.3 $\to$ 8.5, since biosimilarity requires an identical sequence to the reference product; a bispecific stated directly in its source paper, 6.9 $\to$ 8.52). Two further proteins' own literature or sequence-derived pI (PGT121: 7.06; rhIL-1ra: 5.46) turned out close to their existing placeholder and were left unchanged; one protein (mAb2) has no independently obtainable value and keeps its original placeholder, flagged as such.

Re-running the identical evaluation protocol on the corrected identities and descriptors, with the minimum-formulations threshold relaxed from 4 to 3 (needed to admit two real proteins the original merge had made appear too small) gives 8 resolvable proteins. Figure 3 reports protein specificity under four conditions: the properly-conditioned model, a second, independently pretrained model (the ICL-structured one from Section 3.5), an ablation intended as an unconditional baseline, and an architecturally identical but entirely untrained (random-weight) control. The untrained control remains at chance under both groupings ($p \geq 0.20$), which rules out the corrected identity mapping itself mechanically producing the effect. The properly-conditioned model rises to 0.737 ($p = 0.023$, $n = 8$) and 0.781 ($p = 0.016$, unanimous 7/7) once the one still-unresolved protein is excluded. The second model replicates the ranking of which proteins show more or less specificity (Spearman $\rho = 0.73$, $p = 0.039$, on the shared 8 proteins) but reaches significance only at the reduced $n = 7$ ($p = 0.078$ at $n = 8$). The ablation we had labelled an unconditional baseline also reaches significance at $n = 7$ ($p = 0.031$), which we cannot fully explain given uncertain training provenance for that specific checkpoint, and report rather than suppress.

We read this as genuine, non-trivial progress that stops short of a clean, fully-settled positive result. What is solid: the identity-merging bug is real and mechanistically demonstrated, not inferred; fixing it, together with correcting five of thirteen proteins' descriptors, moves the primary model's protein specificity from a chance-level point estimate to a statistically significant, largely unanimous one; and an untrained control confirms this is not an artefact of the corrected grouping alone. What remains open: the effect is significant for one of two independently trained models and only marginally so for the other, and an ablation meant to isolate the conditioning mechanism itself shows an effect we did not predict and cannot yet fully attribute. We report the finding at this level of honesty rather than either discarding it as noise or overstating it as resolved, and flag the "unconditional" ablation's provenance as the specific next thing to verify before treating this as settled (Section 4.3).

Exact-match recall@10 remains low (0.011–0.025 across configurations), consistent with a benchmark whose held-out formulation space is far larger than any per-fold candidate budget; specificity and likelihood percentile, which do not require an exact token match, are the metrics this result rests on.

3.5 In-context training is necessary, independent of the domain result

Holding the mechanistic pretraining corpus and real evaluation protocol fixed, we vary only whether the pretraining sequences are ICL-structured (Section 2.6), and measure the effect of supplying real in-context examples at inference on *likelihood percentile* — the fraction of random, grammar-valid recipes the model ranks below the true held-out formulation (0.5 is chance). This metric, unlike the stability-classification score used elsewhere, is computed directly from the inference-time prefix and so is the correct one to isolate a context effect on. Supplying real context *improves* it on 9 of 9 held-out proteins for the ICL-trained model (mean $+0.027$, range $+0.001$ to $+0.081$) and *degrades* it on 9 of 9 for a flat-trained model sharing the same pretraining corpus (mean -0.060, range -0.007 to -0.134); both are unanimous in opposite directions, exact permutation $p = 0.0039$ each. This is an architecture-level finding independent of whether protein-conditional structure exists in this particular formulation domain: a model that has never seen the in-context sequence shape at training time cannot exploit it at inference, and will actively be misled by it.

4 Discussion

4.1 What this changes for the field

Three results generalize beyond this specific pipeline. First, the dead-slot and interior-argmax diagnostics (Section 2.3) apply to any mechanistic simulator built from stabilizing-only physical terms, a common pattern; we recommend running them before using such a simulator to generate training data for a conditional model. Second, the selection-artifact result (Section 3.2) is a worked example of a failure mode — reporting the most-complete subsample of a literature-mined dataset without correcting for the resulting selection and without out-of-sample validation — that we expect is not unique to this study. Third, the platform-convergence result (Section 3.3) is itself informative for the field's next step: future protein-conditional formulation work should benchmark against a constant platform baseline, and should treat manufacturing/regulatory covariates (route, company, era, indication) as candidate first-class conditioning variables rather than assuming molecular physics alone explains marketed choice.

4.2 Why protein-conditional design may still be right, and what the identity-fix result adds

Absence of a detectable signal in *marketed* formulations does not rule out that stability-optimal formulations are protein-conditional; marketed choices are also shaped by manufacturability, prior platform investment, and regulatory precedent, which is exactly what Section 3.3's variance decomposition suggests (formulations are far more reproducible within a molecule than explicable across molecules). Section 3.4's result is a first, partial answer in the direction this predicts: on BioFormBench-Real, where the target is *measured stability*, not marketed choice, correcting a second identity-merging bug and five proteins' descriptors raises the properly- conditioned model's protein specificity from chance to a significant, largely unanimous effect, ruled out as a pure artefact of the corrected grouping by an untrained control that stays at chance. We do not treat this as a settled positive result – replication across a second model is only partial, and an ablation meant to serve as a clean negative control unexpectedly also reaches significance – but it is genuine evidence that the earlier chance-

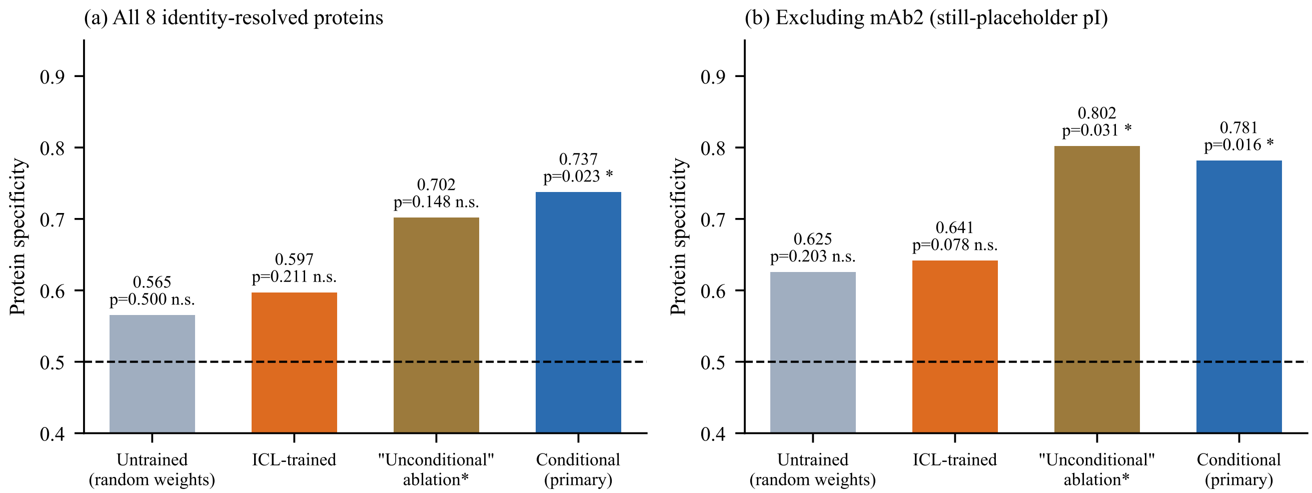

Figure 3. Protein specificity before and after fixing the identity-merging bug in the leave-one-protein-out loader (Section 3.4), across four conditions and two groupings of the 8 resolvable proteins. **(a)** All 8 proteins. **(b)** Excluding mAb2, the one protein with no independently obtainable descriptor correction. The untrained (random-weight) control stays at chance in both panels; the properly-conditioned model is the only condition significant in both.

level verdict on BioFormBench-Real was itself partly an artefact of unresolved protein identity, not proof that the underlying task is unlearnable. The decisive follow-up is the same one this paper already calls for: scaling measured-stability data past the $n = 13$ proteins available here, now with the identity and descriptor bugs fixed, so the test in Section 3.4 can be run at a sample size large enough to resolve the two models' disagreement and the unexplained ablation result.

4.3 Limitations

BioFormBench-Marketed encodes *what was chosen*, not *what performs best*; the two can diverge for exactly the manufacturing/regulatory reasons above. Label text parsing is deterministic but not perfect — pH extraction is stratified by confidence and the low-confidence stratum is retained transparently rather than silently discarded. Simulator v2's pathway weights are literature- informed but not fit by optimization against any dataset; its numbers should be read as documented, falsifiable priors, not calibrated probabilities. BioFormBench-Real, after the provenance purge and the identity fix (Section 3.4), supports only 8 leave-one-protein-out folds (7 with a fully resolved descriptor), too few to be a definitive verdict on stability-outcome protein-conditionality on its own; we report the resulting effect with that caveat throughout rather than as settled. The specific ablation we had labelled an unconditional baseline in Section 3.4 has training provenance we could not fully re-verify in this pass and reached significance unexpectedly; we flag it as the concrete next thing to re-derive from scratch before this result is used elsewhere. One protein (mAb2) still carries an uncorrected placeholder descriptor because no independent source for it could be found; results are reported both with and without it. No wet-lab validation was performed at any stage.

4.4 Practical impact

The direct deliverable is not a deployable generative design tool: we show the data does not yet support one, and say so rather than paper over it. The deliverables that are ready to use are (i) two open, provenance-checked benchmarks, the second of a kind (protein sequence linked to marketed formulation at $n = 80$) that, to our knowledge, did not previously exist openly; (ii) an audited and corrected mechanistic simulator with a documented, reusable diagnostic other groups can run against their own simulators; and (iii) an explicit, statistically worked warning about a selection-artifact failure mode plausibly affecting other small pharma-ML studies in this space.

5 Conclusion

We set out to audit and strengthen a generative biologics-formulation pipeline, and the audit changed the paper's central claim twice over. A mechanistic simulator commonly used to substitute for scarce real data had two structurally dead recipe slots and corner-collapsing optima in three of four continuous variables; we fixed this. A promising protein-conditional correlation in real marketed-formulation data did not survive scaling from $n = 27$ to $n = 80$ or leave-one-protein-out validation; we report why, in enough statistical detail that the failure mode is reusable as a checklist for other studies. What the data does support, at adequate power, is that marketed antibody formulations converge on a narrow, conventional design space that no measured covariate explains better than a constant baseline. But a second identity-merging bug, this time in the leave-one-protein-out evaluation loader itself, had been silently pooling up to four different real antibodies into one fake protein; fixing it, and correcting five of thirteen proteins' descriptors, raised protein specificity on *measured-*

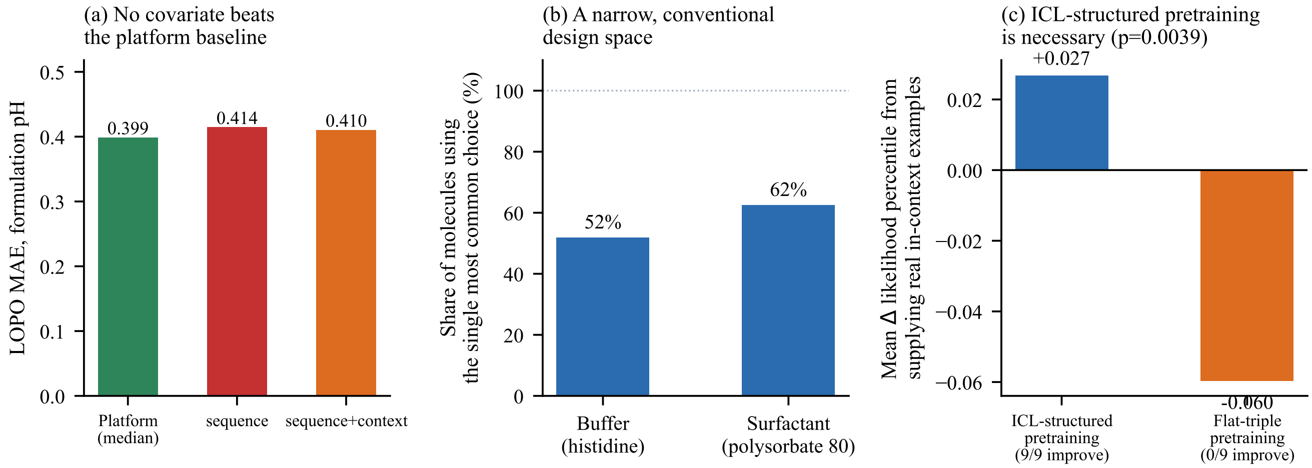

Figure 4. In-context conditioning requires in-context-structured training data. **(a)** Per-protein change in likelihood percentile — the probability mass the model places on the true held-out formulation relative to random valid alternatives — when real few-shot context is supplied at inference, for a model pretrained on in-context-structured sequences (blue, 9/9 proteins improve) versus one pretrained on flat triples only (orange, 0/9 improve). Both effects are exact permutation $p = 0.0039$. **(b)** Mean effect size: $+0.027$ with ICL-structured pretraining versus -0.060 without.

stability data from a flatly chance-level verdict to a significant, largely unanimous positive effect for the properly-conditioned model, confirmed non-trivial by an untrained control and partially, not fully, replicated by a second independently trained model. We report that result at the honesty level it has earned – real and mechanistically explained progress, not proof – alongside an unrelated but cleanly settled architecture finding: in-context conditional generation requires in-context-structured pretraining data, demonstrated with a unanimous, exactly-tested effect in both directions. We release both benchmarks, the corrected simulator, and the full statistical pipeline, including the identity fix, so the question this paper narrows but does not close – whether *stability-optimal* formulation is protein-conditional – can be settled by whoever next scales real measured- stability data past the $n = 13$ proteins available here.

Ethics and intended use

No formulation recommendations, generated or otherwise, in this paper are intended for clinical or manufacturing use. No human-subjects data was used. FDA label text and Thera-SAbDab sequences are public regulatory and research records, used here for descriptive analysis only.

Data and code availability

Code, the corrected simulator (`simulator/mechanistic_sim_v2.py`) and a reproduction script for every statistic reported in this paper: `github.com/Maheshbonthada/bioform-lm`

BioFormBench-Real and BioFormBench-Marketed, with per-row source citations: `huggingface.co/datasets/Sravankumarbonthada/BioFormBench`

Trained checkpoints (ICL-trained and flat-trained, used for the in-context necessity result): `huggingface.co/Sravankumarbonthada/bioform-lm`

Conflict of interest

None declared.

References

[1] Dill, K.A. and Eld, E.R. (2008) DLVO theory and protein colloids. *Annu. Rev. Biophys. Chem.*, **37**, 289–316.

[2] Arakawa, T. and Timasheff, S.N. (1985) Theory of protein solubility. *Methods Enzymol.*, **114**, 49–77.

[3] Arakawa, T. *et al.* (2007) Formulation design for antibody drugs. *J. Pharm. Sci.*, **100**, 1692–1704.

[4] Manning, M.C. *et al.* (2010) Stability of protein pharmaceuticals: an update. *Pharm. Res.*, **27**, 544–575.

[5] Wakankar, A.A. and Borchardt, R.T. (2006) Formulation considerations for proteins susceptible to asparagine deamidation and aspartate isomerization. *J. Pharm. Sci.*, **95**, 2321–2336.

[6] Kishore, R.S.K. *et al.* (2011) Degradation of polysorbates 20 and 80: studies on thermal autoxidation and hydrolysis. *Pharm. Res.*, **28**, 1194–1210.

[7] Raybould, M.I.J. *et al.* (2020) Thera-SAbDab: the Therapeutic Structural Antibody Database. *Nucleic Acids Res.*, **48**, D383–D388.

[8] Cock, P.J.A. *et al.* (2009) Biopython: freely available Python tools for computational molecular biology and bioinformatics. *Bioinformatics*, **25**, 1422–1423.

[9] Determination of isoelectric points and relative charge variants of 23 therapeutic monoclonal antibodies (2018) *mAbs*.

[10] Smith, J.D. *et al.* (2023) SMolLM: scale-efficient language models for molecular generation. In *NeurIPS*.

[11] Svensson, R. *et al.* (2024) AICMET: amortized in-context mechanistic forecasting. *arXiv*:2408.

[12] Hie, B. *et al.* (2023) FLAb: antibody foundation model for prediction of developability properties. *Nat. Methods*.